Nile Nexus at BAREC Shared Task 2026: Enhancing Fine-Grained Arabic Readability Assessment through Ordinal-Aware Transformer Ensembling and Decision Calibration

Sumaiya Zaman, Miftahul Jannat Rishta
Department of Computer Science and Engineering,
Chittagong University of Engineering and Technology, Bangladesh
{u2104110, u2104019}@student.cuet.ac.bd

Abstract

Readability assessment has played a vital role in education, yet most NLP work has focused on English and a few high-resource languages, leaving fine-grained Arabic readability comparatively unexplored. In the BAREC Shared Task 2026, we have addressed sentence-level classification into 19 ordinal readability levels under the Strict Track, fine-tuning four Arabic transformer encoders, MARBERT, CAMeLBERT-MSA, AraELECTRA and AraBERTv2, on the official training split under multiple ordinal-aware losses: soft cross-entropy, squared Earth Mover's Distance, direct ordinal regression and a CORAL-style cumulative-threshold loss. A validation-only greedy ensemble of all resulting models, combined with a learned per-level threshold post-processing step, has raised performance from a best single-model QWK of 0.8341 to a final validation QWK of 0.8507. On the official Blind Test leaderboard, our system has achieved a QWK of 0.848, securing 3rd place in the Strict Track.

1 Introduction

Fine-grained readability assessment has remained largely untested for Arabic, even as it has underpinned curriculum design and content accessibility at scale. The Balanced Arabic Readability Evaluation Corpus (BAREC) has addressed this gap by introducing a large-scale Arabic dataset annotated across 19 ordinal readability levels (Elmadani et al., 2025b, 2026) and the BAREC Shared Task 2026 has provided a benchmark for evaluating models under controlled training conditions.

Sentence-level classification across 19 levels has proven difficult, as adjacent levels often differ only subtly in vocabulary, syntax and semantic complexity and the task's ordinal structure has demanded objectives and evaluation aligned with prediction distance rather than exact-match alone.

We have fine-tuned four Arabic transformer encoders (MARBERT, CAMeLBERT-MSA, AraELECTRA and AraBERTv2) under multiple ordinal-aware objectives, have combined them through a validation-only greedy ensemble optimized for QWK and have introduced learned per-level thresholds to replace conventional rounding.

Our contributions are as follows:

- We have investigated multiple Arabic transformer backbones trained with different ordinal-aware objectives, using loss diversity as an additional ensemble-diversity source.
- We have developed a QWK-optimized greedy ensemble that combines complementary model predictions without using blind-test information.
- We have proposed learned per-level decision thresholds that improve the conversion of continuous scores into ordinal readability classes.

Implementation details are available in our https://github.com/sumaiyaa110/BAREC-Sentence-level-Readability-Assessment—Strict-Track.

2 Background

2.1 Shared Task Description

We have participated in the **Sentence-Level Strict Track** of the BAREC Shared Task 2026 (Elmadani et al., 2026), which focuses on fine-grained Arabic readability assessment. The task is formulated as an ordinal classification problem, where the objective is to predict the readability level of an Arabic sentence among 19 ordered readability categories. Given an input sentence, the system outputs a readability level ranging from 1 (lowest readability level) to 19 (highest readability level).

2.2 Related Work

Arabic readability assessment has evolved from early lexical resources (Al Khalil et al., 2020) to large annotated corpora through recent shared-task

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 179–185
October 28-29, 2026 ©2026 Association for Computational Linguistics

efforts (Alhafni et al., 2024; Elmadani et al., 2025a). The BAREC corpus has extended this progress to fine-grained 19-level labeling (Elmadani et al., 2025b; Habash et al., 2025). BAREC 2025 systems have shown that ensembling pretrained Arabic transformers (Basem et al., 2025) and distance-sensitive losses such as squared Earth Mover's Distance (Abdou, 2025) both improve performance, while others have added morphological features (Elden, 2025) or meta-learned combination (Saeed et al., 2025). !MSA (Basem et al., 2025) has combined multiple Arabic transformer architectures, each paired with its own distinct loss function, within a single ensemble. Our work has extended this direction by training each backbone under multiple different ordinal-aware losses rather than pairing one loss per model, decoupling loss choice from architecture choice as an independent ensemble-diversity axis in the BAREC Shared Task 2026 (Elmadani et al., 2026). Separately, deep ordinal regression has motivated order-aware losses such as CORAL (Cao et al., 2020; Shi et al., 2023) and squared Earth Mover's Distance (Hou et al., 2017). We have built on this by fine-tuning AraBERT (Antoun et al., 2020), CAMeLBERT (Inoue et al., 2021), MARBERT (Abdul-Mageed et al., 2021) and AraELECTRA (Antoun et al., 2021; Clark et al., 2020), chosen for their diverse pretraining corpora and objectives.

3 System Overview

We have formulated Arabic readability assessment on the BAREC Shared Task 2026 as an ordinal classification problem, addressed with a heterogeneous ensemble of Arabic encoders, each fine-tuned under a different ordinal-aware objective, followed by a validation-fitted probability ensemble and a learned per-level decision rule. Figure 1 summarizes this end-to-end pipeline.

3.1 Task Formulation and Data

Given a sentence x, the goal is to predict a readability level $y \in \{1, \ldots, K\}$ with $K = 19$ ordered levels. Standard classification losses have treated all misclassifications as equally costly, whereas QWK has penalized predictions by squared distance from the true level. We have therefore treated the problem as *ordinal* rather than categorical, designing losses, an ensembling strategy and a decision rule that are all QWK-aware.

Letting n_j denote the number of training examples at level j, we have computed inverse-frequency, mean-normalized class weights to counteract the long-tailed label distribution:

$$w_j = \frac{\sum_{k=1}^{K} n_k}{K \cdot n_j}, \qquad j = 1, \ldots, K, \tag{1}$$

used in every classification-style loss below.

3.2 Preprocessing

Input sentences have been tokenized with each backbone's native tokenizer and model-specific normalization (e.g. `ArabertPreprocessor` for AraBERTv2), truncated/padded to 192 subword tokens for tractability on a single T4/P100 GPU, with dynamic padding to a multiple of 8 for mixed-precision efficiency.

3.3 Model Ensemble: Backbones and Ordinal Losses

We have fine-tuned seven backbone/loss combinations across four Arabic encoders and four ordinal-aware objectives, since combining multiple losses on one backbone has contributed more ensemble diversity than backbone choice alone.

3.3.1 Ordinal Soft-Label Loss

For classification-head models (logits $z \in \mathbb{R}^K$), we have combined a class-weighted hard cross-entropy against the true level, a distance-weighted soft cross-entropy against a Gaussian target centered at the true level and a Huber loss on the expected level.

Let $p = \mathrm{softmax}(z)$, true label y and level index vector $\ell = (0, \ldots, K-1)$. The soft target distribution is

$$\tilde{p}_i = \frac{\exp\left(-\frac{(\ell_i - y)^2}{2\sigma^2}\right)}{\sum_k \exp\left(-\frac{(\ell_k - y)^2}{2\sigma^2}\right)}, \qquad \sigma = 1.20. \tag{2}$$

The expected level under the model is $\hat{y} = \sum_i p_i \, \ell_i$. The total loss is

$$\begin{aligned}\mathcal{L}_{\text{ord}} = 0.65\, \mathcal{L}^{w}_{\text{CE}}(z, y) + 0.25\, \mathcal{L}^{w}_{\text{soft-CE}}(p, \tilde{p}) \\ + 0.10\, \mathrm{SmoothL1}(\hat{y}, y),\end{aligned} \tag{3}$$

where $\mathcal{L}^{w}_{\text{CE}}$ and $\mathcal{L}^{w}_{\text{soft-CE}}$ are weighted by the class weights w_y.

3.3.2 Earth Mover's Distance (EMD) Loss

For a second model we have minimized the squared distance between predicted and target cumulative

distribution functions (CDFs) over the ordered levels, penalizing distant mass more heavily than nearby mass:

$$\mathcal{L}_{\text{EMD}} = \sum_{i=1}^{K} \left(F_p(i) - F_y(i)\right)^2, \qquad (4)$$

where

$$F_p(i) = \sum_{k \le i} p_k, \qquad F_y(i) = \mathbb{1}[i \ge y]. \qquad (5)$$

3.3.3 Direct Ordinal Regression

A regression-head model (single scalar output $\hat{y}$) has been trained with mean squared error against the integer level, $\mathcal{L}_{\text{MSE}} = (\hat{y} - y)^2$, its output mapped to a K-way probability distribution via a Gaussian kernel for fair ensembling (Eq. 8).

3.3.4 CORAL-Style Cumulative-Threshold Loss

As a fourth loss family we have used $K - 1$ independent cumulative binary classifiers following the CORAL formulation. For threshold $k \in \{0, \ldots, K-2\}$, the model predicts $\sigma(z_k) \approx P(y > k)$, trained with

$$\mathcal{L}_{\text{CORAL}} = \sum_{k=0}^{K-2} \text{BCE}(z_k, \mathbb{1}[y > k]). \qquad (6)$$

Since the independent linear heads do not guarantee monotonicity of $P(y > k)$ in k, we have projected the predicted cumulative probabilities onto the nearest monotonically non-increasing sequence at inference time before differencing them into class probabilities.

3.3.5 Backbone/Loss Configurations

The final ensemble has combined: MARBERT and AraELECTRA with the ordinal soft loss; CAMeLBERT-MSA with the EMD loss; AraBERTv2 with MSE regression and with the ordinal soft loss; and MARBERT and CAMeLBERT-MSA with the CORAL loss.

3.4 Probability-to-Score Mapping

For any K-way probability vector p, we have defined the continuous expected score

$$s(p) = \sum_{i=1}^{K} p_i \cdot i, \qquad (7)$$

and, for scalar regression outputs $\hat{y}$, mapped back to a smooth K-class distribution using a Gaussian kernel over level distance,

$$p_i \propto \exp\left(-\frac{(i - \hat{y})^2}{2\sigma^2}\right), \qquad \sigma = 1.15. \qquad (8)$$

Discrete level predictions have been obtained by rounding continuous scores, clipped to $[1, K]$, unless the learned decision boundary rule (Section 3.6) has been used instead.

3.5 Validation-Fitted Ensembling

Ensemble weights have been fit on the validation split only, via greedy forward selection over per-model weights $\alpha \in \Delta^{M-1}$: each subsequent model has been blended in (grid search over blend coefficients) only when validation QWK has improved, avoiding the overfitting risk of a dense simplex search.

Probabilities have then been adjusted using per-sample, per-model confidence from normalized entropy,

$$c = 1 - \frac{-\sum_i p_i \log p_i}{\log K}, \qquad (9)$$

re-weighting each model's contribution via exponent $\gamma \in \{0, 0.5, 1.0\}$ and a prior correction toward the training class marginal π with strength $\lambda \in \{0, 0.25, 0.50\}$:

$$p'_i \;\propto\; p_i \cdot \left(\frac{\pi_i}{\bar{p}_i}\right)^{\lambda}, \qquad (10)$$

with the (γ, λ) pair and a global offset (searched over $[-0.80, 0.80]$, step 0.01) chosen to maximize validation QWK. The greedy search has selected four of the seven candidates (MARBERT ordinal, AraELECTRA ordinal, AraBERTv2 regression and AraBERTv2 ordinal) with weights 0.208, 0.150, 0.255 and 0.387 respectively; CAMeLBERT-EMD and both CORAL runs have been excluded (weight 0).

3.6 Decision Calibration and Post-Processing

Rather than one global offset, $K - 1$ decision boundaries have been fit directly in score space against validation QWK via Nelder–Mead, initialized at naive rounding points $\{1.5, \ldots, K - 0.5\}$:

$$\hat{y}_i = \text{clip}\left(1 + \sum_{k=1}^{K-1} \mathbb{1}[s_i > b_k],\ 1,\ K\right), \qquad (11)$$

correcting asymmetric, level-specific skew a single offset cannot address. A competing *off-by-one*

merge rule has also been evaluated, with the higher-QWK rule selected for submission.

4 Experimental Setup

The BAREC dataset has contained Arabic texts from books, articles and educational materials annotated into 19 ordinal readability levels (Elmadani et al., 2025b). We have used the official Train and Development splits for training and validation and have evaluated on the Blind Test set, using no external resources or augmentation per the Strict Track guidelines (Table 1).

Split	Sentences	Words	Levels
Train	54,845	692,029	19
Development	7,310	85,216	19
Blind Test	8,077	95,683	–
Total	70,232	872,928	19

Table 1: Dataset Distribution for BAREC Sentence-Level Readability Classification

4.1 Hyperparameter Settings

Table 2 has shown the training configuration shared across all backbone/loss runs; the model, ordinal loss and random seed used for each individual run has been listed in Table 5 (Appendix B).

Setting	Value
Learning rate	2e-5
Train batch size	8 (16 effective, 2× grad. accum.)
Eval batch size	32
Epochs	5
Max sequence length	192
Weight decay	0.01
Warmup ratio	0.06
Precision	FP16
Optimizer	AdamW

Table 2: Shared training configuration for all backbone/loss runs.

5 Results

Table 3 has shown validation results for all seven backbone/loss combinations, using the four metrics defined in Appendix A.

Table 4 has shown the QWK gain from ensembling and post-processing.

Regression and soft ordinal CE have outperformed CORAL and squared-EMD across all backbones (Table 3), likely because CORAL has needed the monotonicity correction of Section 3.3.4, while squared-EMD's CDF-matching has been less sensitive to adjacent-level errors than direct regression, a pattern corroborated by the greedy search's weighting (Section 3.5).

Run	Loss	QWK	Acc19	Adj. Acc	MAE
arabertv2_mse	Regression	0.8341	0.3681	0.7267	1.1974
arabertv2_ordinal	Soft ordinal CE	0.8303	0.4484	0.7335	1.1471
marbert_ordinal	Soft ordinal CE	0.8211	0.4219	0.7150	1.2152
marbert_coral	CORAL	0.8110	0.3856	0.6932	1.2528
araelectra_ordinal	Soft ordinal CE	0.7795	0.3345	0.6479	1.4508
camelbert_coral	CORAL	0.7727	0.3316	0.6361	1.4235
camelbert_emd	Squared EMD	0.7707	0.3257	0.6317	1.4327

Table 3: Validation performance per backbone/loss combination.

Stage	QWK
Best single model (arabertv2_mse)	0.8341
Greedy ensemble (raw probability blend)	0.8487
+ Per-level threshold optimization (final)	0.8507
Official Blind Test (Strict Track)	0.848

Table 4: QWK across the ensembling/post-processing pipeline and official Blind Test.

6 Conclusion

We have presented an ordinal-aware ensemble for fine-grained Arabic readability assessment on BAREC Shared Task 2026 (Strict Track). Fine-tuning four Arabic encoders under multiple ordinal-aware losses and combining them via validation-only greedy QWK search with learned per-level boundaries has raised QWK from 0.8341 (best single model) to 0.8507 on validation, reaching 0.848 QWK and 3rd place on the Blind Test leaderboard. Loss function has proven a useful ensemble-diversity axis alongside architecture, with soft ordinal cross-entropy and direct regression outperforming CORAL-style and squared-EMD losses.

Limitations

The models have struggled with rare-level and adjacent-level confusion (Appendix C), driven by severe class imbalance at the top of the scale; compute constraints have also restricted us to base-sized encoders and a validation-only ensemble.

Acknowledgement

We have thanked the BAREC Shared Task 2026 organizers for providing the dataset and evaluation framework and the developers of MARBERT,

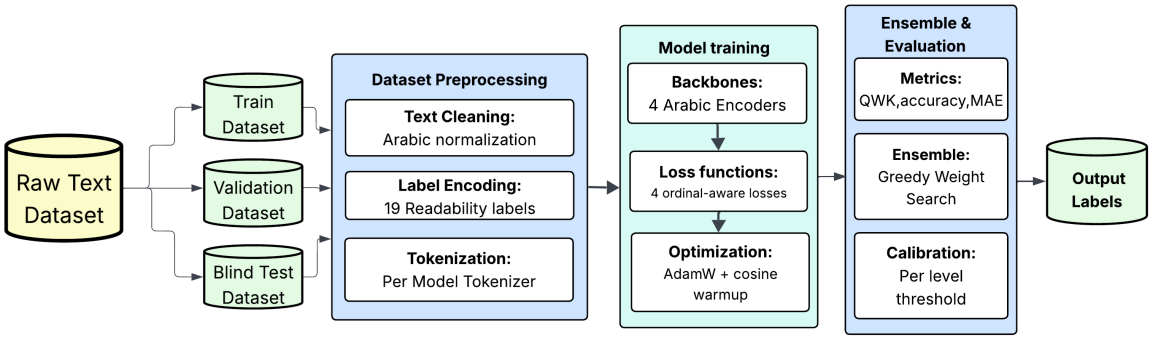


Figure 1: End-to-end pipeline of the heterogeneous 7-model ensemble for fine-grained Arabic readability assessment

CAMeLBERT, AraELECTRA and AraBERTv2, whose open-source pretrained models have made this work possible.

Ethical Statement

All data processing and modeling have adhered to the BAREC Strict Track guidelines to ensure fair, reproducible comparison. Efforts have been made to address class imbalance and level-wise bias in the corpus for fair readability assessment.

References

Ahmed Abdou. 2025. mucAI at BAREC shared task 2025: Towards uncertainty aware Arabic readability assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 312–319, Suzhou, China. Association for Computational Linguistics.

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105, Online. Association for Computational Linguistics.

Muhamed Al Khalil, Nizar Habash, and Zhengyang Jiang. 2020. A large-scale leveled readability lexicon for Standard Arabic. In *Proceedings of the Twelfth Language Resources and Evaluation Conference*, pages 3053–3062, Marseille, France. European Language Resources Association.

Bashar Alhafni, Reem Hazim, Juan David Pineros Liberato, Muhamed Al Khalil, and Nizar Habash. 2024. The SAMER Arabic text simplification corpus. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 16079–16093, Torino, Italia. ELRA and ICCL.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools, with a Shared Task on Offensive Language Detection*, pages 9–15, Marseille, France. European Language Resources Association.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2021. AraELECTRA: Pre-training text discriminators for Arabic language understanding. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 191–195, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Mohamed Basem, Mohamed Younes, Seif Ahmed, and Abdelrahman Moustafa. 2025. !MSA at BAREC shared task 2025: Ensembling Arabic transformers for readability assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 297–305, Suzhou, China. Association for Computational Linguistics.

Wenzhi Cao, Vahid Mirjalili, and Sebastian Raschka. 2020. Rank consistent ordinal regression for neural networks with application to age estimation. *Pattern Recognition Letters*, 140:325–331.

Kevin Clark, Minh-Thang Luong, Quoc V. Le, and Christopher D. Manning. 2020. ELECTRA: Pre-training text encoders as discriminators rather than generators. In *International Conference on Learning Representations*.

Fatimah Mohamed Emad Elden. 2025. MorphoArabia at BAREC 2025 shared task: A hybrid architecture with morphological analysis for Arabic readability assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 286–296, Suzhou, China. Association for Computational Linguistics.

Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2025a. BAREC shared task 2025 on Arabic readability assessment. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 239–252, Suzhou, China. Association for Computational Linguistics.

Khalid N. Elmadani, Bashar Alhafni, Hanada Taha, and Nizar Habash. 2026. BAREC-ST-2026: The second shared task on Arabic readability assessment. In

Proceedings of the Fourth Arabic Natural Language Processing Conference, Budapest, Hungary. Association for Computational Linguistics.

Khalid N. Elmadani, Nizar Habash, and Hanada Taha-Thomure. 2025b. A large and balanced corpus for fine-grained arabic readability assessment. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 16376–16400, Vienna, Austria. Association for Computational Linguistics.

Nizar Habash, Hanada Taha-Thomure, Khalid N. Elmadani, Zeina Zeino, and Abdallah Abushmaes. 2025. Guidelines for fine-grained sentence-level arabic readability annotation. In *Proceedings of the 19th Linguistic Annotation Workshop (LAW-XIX-2025)*, pages 359–376, Vienna, Austria. Association for Computational Linguistics.

Le Hou, Chen-Ping Yu, and Dimitris Samaras. 2017. Squared earth mover's distance-based loss for training deep neural networks. *arXiv preprint arXiv:1611.05916*.

Go Inoue, Bashar Alhafni, Nurpeiis Baimukan, Houda Bouamor, and Nizar Habash. 2021. The interplay of variant, size, and task type in Arabic pre-trained language models. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 92–104, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Mostafa Saeed, Rana Waly, and Abdelaziz Ashraf Hussein. 2025. AMAR at BAREC shared task 2025: Arabic meta-learner for assessing readability. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 320–330, Suzhou, China. Association for Computational Linguistics.

Xintong Shi, Wenzhi Cao, and Sebastian Raschka. 2023. Deep neural networks for rank-consistent ordinal regression based on conditional probabilities. *Pattern Analysis and Applications*, 26:941–955.

A Evaluation Metrics

Model selection and validation reporting have used four metrics: Quadratic Weighted Kappa (QWK, the shared task's primary metric), 19-way exact accuracy, adjacent accuracy (proportion of predictions within ± 1 level of the gold label) and mean absolute error (MAE) in levels.

B Reproducibility

All random seeds (Python, NumPy, PyTorch and the Hugging Face `Trainer` seed) have been fixed. Each backbone/loss run has used a distinct seed to promote ensemble diversity while keeping every individual run reproducible. Table 5 lists the backbone, ordinal loss and seed used for each run.

Run	Backbone/Loss	Seed
marbert_ordinal	MARBERT / Soft CE	42
camelbert_emd	CAMeLBERT-MSA / EMD	52
araelectra_ordinal	AraELECTRA / Soft CE	62
arabertv2_mse	AraBERTv2 / MSE	72
marbert_coral	MARBERT / CORAL	142
camelbert_coral	CAMeLBERT-MSA / CORAL	152
arabertv2_ordinal	AraBERTv2 / Soft CE	82

Table 5: Backbone, ordinal loss and seed used for each run in the ensemble.

C Error Analysis

We have conducted quantitative and qualitative error analyses of the best-performing (post-processed ensemble) system on the Development set.

Quantitative Analysis: Figure 2 has grouped predictions into five readability bands. Accuracy has been 0.79, 0.76, and 0.79 for bands 1–4, 5–8 and 13–16, respectively, dropping to 0.69 for band 9–12 and 0.56 for band 17–19, where 0.42 of sentences have been confused with the neighboring 13–16 band. Confusion beyond the adjacent band has been negligible ($\leq$0.03) everywhere, confirming that errors have been localized to neighboring levels. The weak 17–19 performance has reflected severe class imbalance at the top of the scale (levels 17, 18 and 19 have only 364, 67 and 85 training sentences).

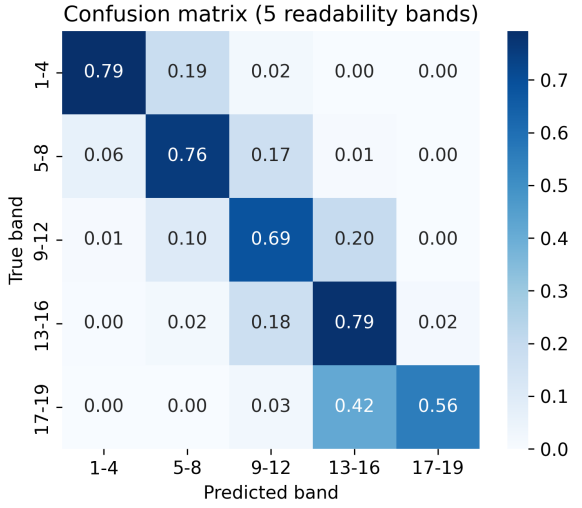


Figure 2: Confusion matrix grouped into five readability bands.

Calibration Analysis: Figure 3 has presented a reliability diagram for the final post-processed ensemble on the Development set, binning predictions by confidence against exact-match accuracy.

The model has shown mild overconfidence below a confidence of roughly 0.5 and underconfidence above it, crossing the perfect-calibration line near the midpoint. This has aligned with the per-level threshold optimization in Section 3.6, which has corrected level-specific decision skew rather than a uniform confidence shift.

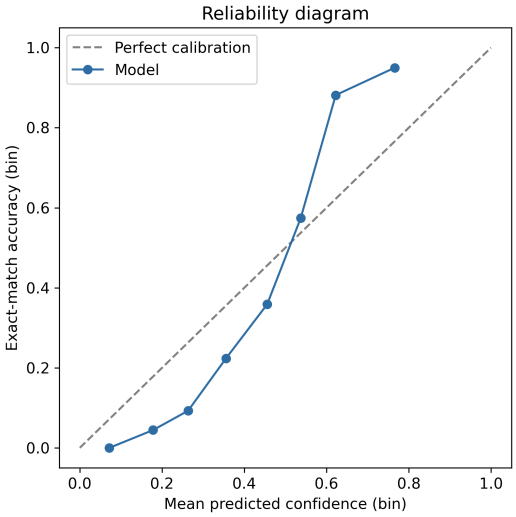


Figure 3: Reliability diagram: mean predicted confidence vs. exact-match accuracy per bin, Development set.

Decision Boundary Shift: Figure 4 has compared the $K-1$ learned decision boundaries against the naive rounding points $\{1.5, \ldots, K-0.5\}$. The learned boundaries have sat consistently to the right of the naive thresholds, with the gap widening toward levels 17–19. This rightward shift has indicated that the raw ensemble scores under-predict level for a given true label, an effect most pronounced where severe class imbalance has left the model least confident, corroborating the weak 17–19 band performance in Figure 2.

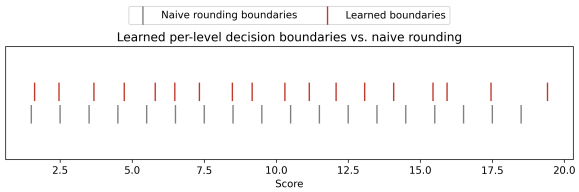


Figure 4: Learned per-level decision boundaries (Section 3.6) compared to naive integer-rounding thresholds, Development set.